Graduation Project Documentation

Detecting Autism Spectrum Disorder via Facial Expressions and FMRI

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APPROVAL PAGE

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Finally, we extend our appreciation to all the contributors and communities involved in creating publicly available datasets like the ABIDE dataset and the Kaggle Autism Image Data. Their contributions are pivotal in advancing research in autism detection and early diagnosis.

DECLARATION

We, the undersigned, declare that this graduation project, titled “Detecting Autism via Facial Expressions and FMRI (ABIDE),” is the original work conducted and compiled by us under the supervision of Dr. Dina Abdelhafiz Hussein. All the data, results, and analyses included in this report are truthful and derived from primary research and secondary sources as appropriately cited.

This project has been undertaken as part of the requirements for the completion of the Bachelor of Science degree in Computer and Data Science at Alexandria University, Egypt, for the academic year 2024–2025.

We confirm that this report has not been submitted, in part or whole, to any other institution or for any other purpose.

Date: 2024-2025

Contents

1 Chapter 1: Introduction 1
1.1 The Global Prevalence and Impact of ASD 1
1.2 Challenges in Current Diagnostic Methods 2
1.3 Rationale for the Dual Approach in This Project 2
1.4 Relevance to Healthcare and Society 3
1.5 Objectives 3
1.6 Deliverables 4
1.7 Roles of Each Member 4
1.8 Schedule 7
1.9 Overview 8
2 Chapter 2: Literature Review 9
3 Chapter 3: Datasets 10
3.1 Facial Expression Dataset 10
3.2 FMRI Dataset 11
3.3 Preprocessing 14
3.3.1 FMRI Data Preprocessing 14
3.3.2 FMRI Data Preprocessing 14
4 Chapter 4: Deep Learning Models 15
4.1 General Description of the Project 15
4.2 Facial Expression Models 15
4.2.1 ResNet-50: 15
4.2.2 EfficientNet-B0: 17
4.2.3 VGG16 18
4.3 FMRI Models 20
4.3.1 3D CNNs 20
4.3.2 Graph Neural Networks (GNNs): 21
5 Chapter 5 : Methods and Materials 22
5.1 Software Tools and Libraries 23
5.2 Hardware Specifications 24

5.3 Facial Expression Analysis Using VGG16 24
5.3.1 Dataset Preparation 24
5.3.2 Model Architecture 25
5.3.3 Training Process 27
5.3.4 Evaluation and Results 28
5.3.5 Discussion 29
5.4 FMRI Data Analysis Using 3D CNN 30
5.4.1 Dataset Preparation 30
5.4.2 Model Architecture 31
5.4.3 Training the Model 32
5.4.4 Evaluation and Results 32
5.4.5 Discussion 33
6 Chapter 6: UX/UI and Platform Design 34
7 Chapter 7: Conclusion and Recommendations 42
8 References 44

LIST OF FIGURES

Figure 1 Shows the ASD statistics in 2023 [1] 1
Figure 2 Resnet-50 architecture [11] 16
Figure 3 Efficient net diagram [11] 17
Figure 4 Block of digram VGG16 [12] 19
Figure 5 Facial Expressions Data Preview 25
Figure 6 Facial Expressions Data Preview 25
Figure 7 Facial Expressions Data Preview 25
Figure 8 Shows the VGG16 Architecture used [13] 25
Figure 9 Shows the model summary 26
Figure 10 Shows training and validation Accuracy 27
Figure 11 Shows training and validation Loss 27
Figure 12 Shows the evaluation metrics used for the model 28
Figure 13 Shows the Confusion Matrix for the model 28
Figure 14 Shows Data Preview 1 30
Figure 15 Shows Data Preview 2 30
Figure 16 Shows Data Preview 3 30
Figure 17 3D CNN Model Architicture 31
Figure 18 Shows the Model's Training and Validation Accuracy 32
Figure 19 UI Splash Page 34
Figure 20 UI Sign Up Page 35
Figure 21 UI Sign In Page 36
Figure 22 UI Home Page 37
Figure 23 UI Doctors Page 38
Figure 24 UI Appointment Page 39
Figure 25 UI Scan Page 40

LIST OF TABLES

Table 1 Shows the time frame of each task and its status during first semester. 7
Table 2 Shows the time frame of each task and its status during second semester. 7
Table 3 Shows the datasets we worked with in details of each. 13

ABBREVIATIONS

The next list describes several abbreviations that will be later used within the body of this documentation.

ABIDE	Autism Brain Imaging Data Exchange
AD	Autism Disease
ADOS	Autism Diagnostic Observation Schedule
ADI-R	Autism Diagnostic Interview-Revised
AI	Artificial Intelligence
ALFF	Amplitude of Low-Frequency Fluctuations
API	Application Programming Interface
APP	Application
ASD	Autism Spectrum Disorder
AST	Asthma Symptom Tracker
AUC	Area Under the Curve
CDC	Centers for Disease Control and Prevention
CNN	Convolutional Neural Networks
C-PAC	Configurable Pipeline for the Analysis of Connectomes
CORS	Cross-Origin Resource Sharing
DL	Deep Learning
DT	Decision Tree
EEG	Electroencephalography
FFT	Fast Fourier Transform
FN	False Negative
FP	False Positive
FMRI	Functional Magnetic Resonance Imaging
GBD	Global Burden of Disease
GNN	Graph Neural Networks

GPS	Global Positioning System
HIPAA	Health Insurance Portability and Accountability Act
HTTP	Hypertext Transfer Protocol
ICBHI	International Conference on Biomedical and Health Informatics
IOT	Internet of Things
JSON	JavaScript Object Notation
KAUH	King Abdullah University Hospital
KNN	k-Nearest Neighbors
LDC	Lung Disease Classification
LRTI	Lower Respiratory Tract Infection
MFCC	Mel-Frequency Cepstral Coefficients
ML	Machine Learning
MNI	Montreal Neurological Institute
PCA	Principal Component Analysis
PEF	Peak Expiratory Flow
PFM	Peak Flow Meter
PSO	Particle Swarm Optimization
ReLU	Rectified Linear Unit
RF	Random Forest
ROC	Receiver Operating Characteristic
ROI	Region of Interest
SEN	Sensitivity
SMOTE	Synthetic Minority Oversampling Technique
SOS	Save Our Ship
SPE	Specificity
SVM	Support Vector Machine
TN	True Negative
TP	True Positive
UI	User Interface
ViT	Vision Transformer

WHO World Health Organization

Abstract

Autism Spectrum Disorder (ASD) is a complex developmental condition characterized by challenges in social interaction, communication, and behavioral adaptability. The rising prevalence of ASD globally emphasizes the urgent need for reliable, efficient, and early diagnostic tools to support timely intervention and improve developmental outcomes. This project proposes a comprehensive and innovative approach to ASD detection by employing advanced deep learning techniques across two complementary domains:

- **Facial Expression Analysis**: Leveraging the Kaggle Autism Image Dataset, the project employs the VGG16 model, a 16-layer convolutional neural network, to classify subtle facial characteristics associated with ASD. The model achieved an accuracy of 73%, demonstrating its effectiveness in capturing facial patterns indicative of the disorder.

- **Functional Magnetic Resonance Imaging (FMRI) Analysis**: Using the ABIDE dataset, preprocessed with the Configurable Pipeline for the Analysis of Connectomes (C-PAC), the project applies 3D CNNs to analyze brain activity patterns. The model achieved an accuracy of 68%, highlighting its ability to detect neurological biomarkers of ASD.

The methodologies include rigorous preprocessing, state-of-the-art model implementation, and performance evaluation using metrics such as accuracy, precision, recall, and F1-score. A comparative study highlights the strengths and limitations of each approach. The project also designs a user-friendly platform to bridge the gap between research and clinical use, offering a scalable and non-invasive tool for early ASD detection.

Future Directions: Expanding datasets, exploring advanced architectures, implementing the work into a more user friendly platform to be used by both doctors and people

Chapter 1: Introduction

Autism Spectrum Disorder (ASD) is a complex neurodevelopmental condition that affects millions of individuals worldwide. Despite extensive research, the exact causes of autism remain elusive, with a mix of genetic, environmental, and neurological factors thought to play significant roles. Early diagnosis and intervention are critical in improving the quality of life and developmental outcomes for individuals with autism. However, current diagnostic tools often rely on subjective evaluations and behavioral assessments, which can delay timely intervention.

1.1 The Global Prevalence and Impact of ASD

The prevalence of ASD has steadily increased over the last few decades. According to the Centers for Disease Control and Prevention (CDC), approximately 1 in 54 children in the United States is diagnosed with autism. This growing prevalence highlights the need for innovative diagnostic tools that can complement traditional methods. The global impact of ASD extends beyond the individuals diagnosed, affecting families, caregivers, educators, and healthcare systems. The economic burden associated with autism, including healthcare costs, educational support, and lost productivity, underscores the necessity of early and accurate diagnosis.

Shown in figure 1 some of the symptoms and the statistics about the ASD.

Figure 1 Shows the ASD statistics in 2023 [1]

1.2 Challenges in Current Diagnostic Methods

Traditional diagnostic methods for ASD, such as the Autism Diagnostic Observation Schedule (ADOS) and the Autism Diagnostic Interview-Revised (ADI-R), are time-intensive and require trained professionals. These tools primarily rely on subjective evaluations of behavioral symptoms, which can vary significantly between individuals. Additionally, cultural and linguistic differences may influence the assessment process, potentially leading to underdiagnoses or misdiagnosis in certain populations.

Early behavioral markers of autism, such as delayed speech, lack of eye contact, or repetitive behaviors, may not always be apparent in young children, further complicating the diagnostic process. The reliance on observable behaviors often results in a diagnosis occurring after critical developmental periods, limiting the effectiveness of early intervention programs.

1.3 Rationale for the Dual Approach in This Project

This project explores two complementary approaches for ASD diagnosis: facial expression analysis and functional magnetic resonance imaging (FMRI) analysis. Each method offers unique insights into autism's underlying characteristics.

- Facial Expression Analysis: Research suggests that individuals with ASD exhibit atypical facial expressions, gaze patterns, and emotional responses. Machine learning models, particularly convolutional neural networks (CNNs), are well-suited for analyzing these subtle differences. By leveraging image-based datasets, this approach aims to provide a non-invasive and accessible diagnostic tool.
- FMRI Analysis: Neuroimaging studies have identified differences in brain connectivity and activity in individuals with autism. FMRI data provides a window into the brain's functional architecture, revealing patterns associated with social cognition, sensory processing, and executive functioning. This project employs advanced preprocessing

techniques and deep learning algorithms to extract meaningful biomarkers from FMRI scans.

1.4 Relevance to Healthcare and Society

The proposed methods have the potential to revolutionize autism diagnosis, addressing several key challenges in current practices. Early detection facilitated by these tools can improve access to intervention services, enhancing developmental outcomes and reducing long-term societal costs. Additionally, the integration of AI-driven diagnostics into clinical workflows aligns with broader trends in precision medicine, where treatments are tailored to individual needs based on data-driven insights.

Beyond healthcare, this project contributes to the growing body of research on the applications of AI in neurodevelopmental disorders. By demonstrating the feasibility and efficacy of these methods, the project paves the way for future innovations in ASD diagnosis and support.

1.5 Objectives

The primary objectives of this project are to address the growing need for early and accurate diagnostic tools for Autism Spectrum Disorder (ASD). By employing advanced machine learning techniques across two distinct methodologies-facial expression analysis and neuroimaging data analysis

this project seeks to:

- Develop robust machine learning models to classify ASD using facial expression datasets and FMRI data.
- Conduct a comparative analysis of various deep learning architectures to identify the most effective approaches for ASD detection.
- Build an integrated, user-friendly diagnostic platform that combines insights from both methodologies.
- Disseminate findings through academic publications and contribute to the growing body of research on AI-based healthcare solutions.

1.6 Deliverables

The deliverables for this project encompass a wide range of outcomes that reflect its comprehensive approach:

1. **Machine Learning Models**: Trained and validated models for both facial expression and neuroimaging data analysis.
2. **Performance Metrics**: Detailed performance evaluations, including accuracy, precision, recall, and F1-scores for each model.
3. **Integrated Platform**: A fully operational diagnostic platform combining the results of facial expression and neuroimaging analysis.
4. **Project Documentation**: Comprehensive reports detailing the methodologies, findings, and recommendations.
5. **Published Research**: Submission of project findings to academic journals and presentations at relevant conferences.

1.7 Roles of Each Member

- **In the First Semester**

Our team divided the workload strategically among its members to ensure effective progress during the first semester:

- **Emad Reda El-Sayed** performed comparative analyses of preliminary model performances, ensuring that improvements were data-driven and evidence-based.
- **Tarek Mahmoud Zaki** focused on preprocessing the neuroimaging data. He implemented the Configurable Pipeline for the Analysis of Connectomes (C-PAC), handling complex processes such as voxel standardization and feature extraction.
- **Al-Moatasem Mohammed Mofadel** focused on preparing the interim presentations and documenting findings from the first semester.

- **Abdelrahman Waleed** specialized in designing deep learning architectures for neuroimaging data. He experimented with 3D CNNs and other advanced techniques to analyze brain activity patterns.
- **Yusuf Waleed** conducted a thorough literature review to identify existing research and curated datasets for both methods, providing the foundation for the project's theoretical framework.
- **Moamen Ahmed Mohamed** led the preprocessing of facial expression datasets. He ensured the quality and consistency of the images by applying resizing, augmentation, and normalization techniques to prepare them for model training.
- **Shahd Usama** managed documentation, tracked team progress, and ensured alignment with the project timeline. She also contributed to the interim report.
- **Shahd Hisham** worked on developing and training initial CNN models for facial expression analysis, using transfer learning and data augmentation to improve accuracy.
- **Menna Shafik** designed the initial user interface, creating a prototype for the platform and setting the groundwork for the second semester's development.

- ## In the Second Semester

During the second semester, the focus shifted to refining models, integrating components, and finalizing the platform:

- **Emad Reda El-Sayed** lead the integration of all developed models into the diagnostic platform. Ensuring that facial expression and neuroimaging models worked cohesively.
- **Tarek Mahmoud Zaki** Conducts real-time testing and evaluation of the platform's performance, identifying and resolving any technical issues.
- **Al-Moatasem Mohammed Mofadel** Refine the facial expression models, incorporating feedback from testing phases to enhance accuracy and robustness.
- **Abdelrahman Waleed** Optimizing the neuroimaging models, applying hyperparameter tuning and experimenting with alternative architectures.

- **Yusuf Waleed** performing the final comparative analysis, consolidating findings to evaluate the overall effectiveness of the models.
- **Momen Ahmed Mohamed** working on platform optimization, focusing on scalability and ensuring a smooth user experience.
- **Shahd Usama** finalizing the project documentation, integrating all findings into a comprehensive report.
- **Shahd Hisham** preparing the research publications, ensuring the dissemination of results to academic and professional audiences.
- **Menna Shafik** polishing the platform's design, ensuring it was user-friendly and visually appealing. She also supported the development of user manuals.

1.8 Schedule

Shown in Table 1 and Table 2 the schedule we worked with during the 1st semester, and the schedule we will be working with during the 2nd semester.

Table 1 Shows the time frame of each task and its status during first semester.

Time Frame	Task Name	Status
Week 1-4	Collect datasets for facial expression and neuroimaging analysis.	Completed
	Preprocess datasets, including resizing images and preparing FMRI pipelines using C-PAC.	Completed
Week 5-8	Develop initial models for facial expression analysis using CNNs.	Completed
	Design initial architectures for neuroimaging analysis.	Completed
Week 9-12	Evaluate model performance and identify areas for improvement.	Completed
	Submit an intermediate report summarizing progress and findings.	Completed

Table 2 Shows the time frame of each task and its status during second semester.

Time Frame	Task Name	Status
Week 1-4	Refine models based on first semester evaluation results.	Not Completed
	Begin integrating facial expression and neuroimaging models into a unified platform.	Not Completed
Week 5-8	Conduct real-time testing of the integrated platform.	Not Completed
	Optimize platform performance for scalability and accessibility.	Not Completed
Week 9-12	Prepare the final presentation and submit comprehensive documentation.	Not Completed
	Develop and submit research publications based on project outcomes.	Not Completed

1.9 Overview

This report is structured into several chapters, each addressing key aspects of the project:

- **Chapter 1:** Introduction outlines Autism Spectrum Disorder (ASD), its prevalence, challenges in diagnosis, and the role of technology in advancing early detection, the objectives, deliverables, roles, schedule, and an overview of the project's significance.
- **Chapter 2:** Literature Review examines existing research and provides context for the methodologies used.
- **Chapter 3:** Datasets describes the datasets utilized, including their sources and preprocessing.
- **Chapter 4:** Deep learning models details the architectures and algorithms employed.
- **Chapter 5:** Materials and methods provides a comprehensive explanation of the experimental process and models.
- **Chapter 6:** Design and Implementation focuses on the platform's development and functionality.
- **Chapter 7:** Conclusion and Recommendations summarizes the outcomes and suggests future directions..

Chapter 2: Literature Review

In this chapter, we discuss the references used in our project, examining their methodologies, results, and relevance. These studies highlight the application of machine learning, CNNs, and deep learning techniques in ASD detection using facial expression and FMRI data.

Liu et al. (2016) explored the use of Convolutional Neural Networks (CNNs) for analyzing facial expression videos, achieving 89% accuracy in distinguishing autistic traits and emphasizing the importance of dynamic analysis for capturing temporal changes indicative of ASD [2].

Almufareh et al. (2020) further advanced this by investigating Vision Transformers (ViTs) and CNNs, achieving 92% accuracy in classifying facial images of autistic and non-autistic children, highlighting the potential of advanced architectures for ASD detection [3].

Ahmed et al. (2019) demonstrated the effectiveness of transfer learning using VGG16 on a limited dataset, achieving 87% accuracy through data augmentation techniques [4].

In neuroimaging, Heinsfeld et al. (2018) used 3D CNNs to analyze resting-state FMRI data from the ABIDE dataset, achieving 70% accuracy and laying the groundwork for deep learning in ASD research [5].

Di Martino et al. (2014) introduced preprocessing strategies like ALFF and filt_noglobal to improve FMRI data reliability, providing a standardized framework for future studies [6].

Zhou et al. (2020) explored multimodal deep learning by combining structural MRI, FMRI, and behavioral data, showing that late fusion techniques improved diagnostic accuracy [8, 9].

Similarly, Hazlett et al. (2017) integrated FMRI and eye-tracking data, underscoring the potential of combining neuroimaging and behavioral data to enhance ASD diagnosis [10].

These studies collectively highlight the evolving role of advanced machine learning techniques and multimodal approaches in improving ASD detection and diagnosis.

Chapter 3: Datasets

3.1 Facial Expression Dataset

The **Facial Expression Dataset** is a critical component of your project, as it provides the visual data necessary for training machine learning models to detect ASD based on facial characteristics.

- **Source**: The dataset is sourced from Kaggle Autism Image Data [11] , a publicly available repository that contains labeled images of individuals with and without ASD.

- **Description**:
  - The dataset contains 3000 labeled images, divided into two categories:
  - Autistic: Images of individuals diagnosed with Autism Spectrum Disorder.
  - Non-Autistic: Images of neurotypical individuals (control group).
  - The age range of participants spans from children to adolescents, ensuring that the dataset captures a wide range of developmental stages.
  - The images are captured under varied lighting conditions, poses, and emotional expressions. This diversity improves the generalizability of the trained models, as they are exposed to a wide range of real-world scenarios.

- **Key Features:**
  - Labeling: Each image is clearly labeled as "Autistic" or "Non-Autistic," making it suitable for supervised learning tasks.
  - Diversity: The dataset includes images with different facial expressions (e.g., happy, sad, neutral) and poses (e.g., frontal, side profiles), which helps the model learn robust features.
  - Preprocessing: The dataset may require preprocessing steps such as facial alignment, normalization, and augmentation to ensure consistency and improve model performance.

- **Applications:**
  - The dataset is used to train Convolutional Neural Networks (CNNs), Vision Transformers (ViTs), and other deep learning models to classify individuals as autistic or non-autistic based on facial features.
  - The models can detect subtle differences in facial expressions, gaze patterns, and emotional responses that are characteristic of ASD.

3.2 FMRI Dataset

The Autism Brain Imaging Data Exchange (ABIDE) dataset is a cornerstone of your project, providing neuroimaging data that allows for the analysis of brain activity patterns in individuals with ASD.

- **Source:** The dataset is sourced from the ABIDE initiative, a collaborative effort that aggregates FMRI data from multiple research sites worldwide. It is one of the largest and most widely used datasets for ASD research.

- **Description**:
  - Includes preprocessed FMRI scans from over 1000 individuals, with balanced representation of ASD and control participants.
    - Autistic Individuals: 601 participants diagnosed with ASD.
    - Control Participants: 434 neurotypical individuals.
  - Scans are preprocessed with the Configurable Pipeline for the Analysis of Connectomes (C-PAC) using the following configurations:
    - **Strategy**: filt_noglobal (to mitigate global signal confounds).
    - **Derivative**: is extracted for frequency-domain feature analysis. ALFF measures the strength of low-frequency oscillations in the brain, which are often altered in individuals with ASD.
  - The dataset also includes phenotypic information (e.g., age, gender, diagnosis) stored in a CSV file (Phenotypic_V1_0b_preprocessed1.csv), which can be used for additional analysis and model training.

- **Technical Details:**
  - **Image Format:** The FMRI scans are stored in NIfTI (.nii.gz) format, a common format for neuroimaging data. Each scan has a voxel resolution of 61x73x61, representing the spatial dimensions of the brain.
  - **Preprocessing:**
    - **C-PAC Pipeline**: The FMRI data undergoes several preprocessing steps, including motion correction, slice timing correction, and normalization to a standard brain template (e.g., MNI space).
    - **Feature Extraction:** In addition to ALFF, other features such as functional connectivity and region-of-interest (ROI) analysis can be extracted from the FMRI data.
- **Applications:**
  - The dataset is used to train 3D Convolutional Neural Networks (3D CNNs), Graph Neural Networks (GNNs), and autoencoders to analyze brain activity patterns associated with ASD.
  - The models can identify differences in brain connectivity, activity, and structure that are indicative of ASD, providing a neurological basis for diagnosis.

As shown in table 3, The summary of both datasets with the Source, Dataype, Sample Size, Preprocessing, Key Features, Model Application and Output for both datasets used.

Table 3 Shows the datasets we worked with in details of each.

Aspect	Facial Expression Dataset	FMRI Dataset (ABIDE)
Source	Kaggle Autism Image Data	Autism Brain Imaging Data Exchange (ABIDE)
Datatype	2D images (RGB)	3D FMRI scans (NIfTI format)
Sample Size	3000 images (Autistic and Non-Autistic)	1035 FMRI scans (601 Autistic, 434 Control)
Preprocessing	Facial alignment, normalization, augmentation	C-PAC pipeline (filt_noglobal, ALFF extraction)
Key Features	Facial expressions, poses, lighting conditions	Brain connectivity, ALFF, functional connectivity
Model Applications	VGG16,Mobile Net, Resnet	3DCNNs,GNNs, Autoencoders
Output	Binary classification (Autistic vs. Non-Autistic)	Binary classification (Autistic vs. Control)

3.3 Preprocessing

3.3.1FMRI Data Preprocessing

1. **Augmentation**:
    - Random cropping, flipping, and rotation.
    - Adding Gaussian noise to simulate real-world imperfections.
2. **Normalization**:
    - Scale pixel intensity to [0, 1] for consistent input.
3. **Facial Alignment**:
    - Use Dlib or OpenCV to detect and align faces based on key points (e.g., eyes, nose, and mouth).
4. **Feature Extraction**:
    - Convert images to grayscale for texture-focused analysis.
    - Extract Histogram of Oriented Gradients (HOG) or use CNN feature maps.

3.3.2 FMRI Data Preprocessing

1. **Pipeline:** C-PAC.
2. **Strategy:** filt_noglobal to remove global signals.
3. **Derivative:** ALFF extraction for frequency analysis.
4. **Dimensionality Reduction**:
    1. Principal Component Analysis (PCA) to identify dominant patterns.
    2. Autoencoders for nonlinear feature extraction.
5. **Region of Interest (ROI) Extraction**:
    1. Use brain atlases (e.g., AAL or Harvard-Oxford) to segment the brain into functional regions.
6. **Time-Series Analysis**:
    1. Calculate ALFF and functional connectivity metrics.
    2. Extract graph-based features using Graph Neural Networks (GNN).
7. **Balancing and Augmentation**:
    1. Use oversampling techniques like SMOTE to address class imbalances.

Chapter 4: Deep Learning Models

4.1 General Description of the Project

The goal of this chapter is to outline the architecture and configurations of the deep learning models selected for the ASD detection project. Each model is tailored to exploit the unique properties of its respective data modality: facial expressions and FMRI scans. The models are chosen based on their ability to handle high-dimensional data, extract meaningful features, and generalize well on unseen data.

For both facial expressions and FMRI, we consider three models each, selected for their proven effectiveness in similar domains.

Models Used for Both:

Facial Expressions: VGG16, ResNet-50 and EfficientNet-B0.

FMRI Scans: 3D CNNs, GNN.

4.2 Facial Expression Models

The models used in the facial expressions project are as follows:

4.2.1 ResNet-50:

Description: A deep residual network designed to address the vanishing gradient problem in very deep networks. ResNet-50 utilizes skip connections to allow gradients to flow through shortcut connections, ensuring robust feature extraction.

- **Hyperparameters**:
  - Learning rate: 0.001 with decay.
  - Optimizer: Adam.
  - Batch size: 32.
  - Data augmentation: Enabled.

- **Architecture**:

  - 50 layers with residual blocks.
  - Feature extraction followed by global average pooling.
  - Fully connected layer for binary classification (Autistic vs. Non-Autistic).

  Shown in Figure 2 a graphical modeling of Resnet-50 Architecture.

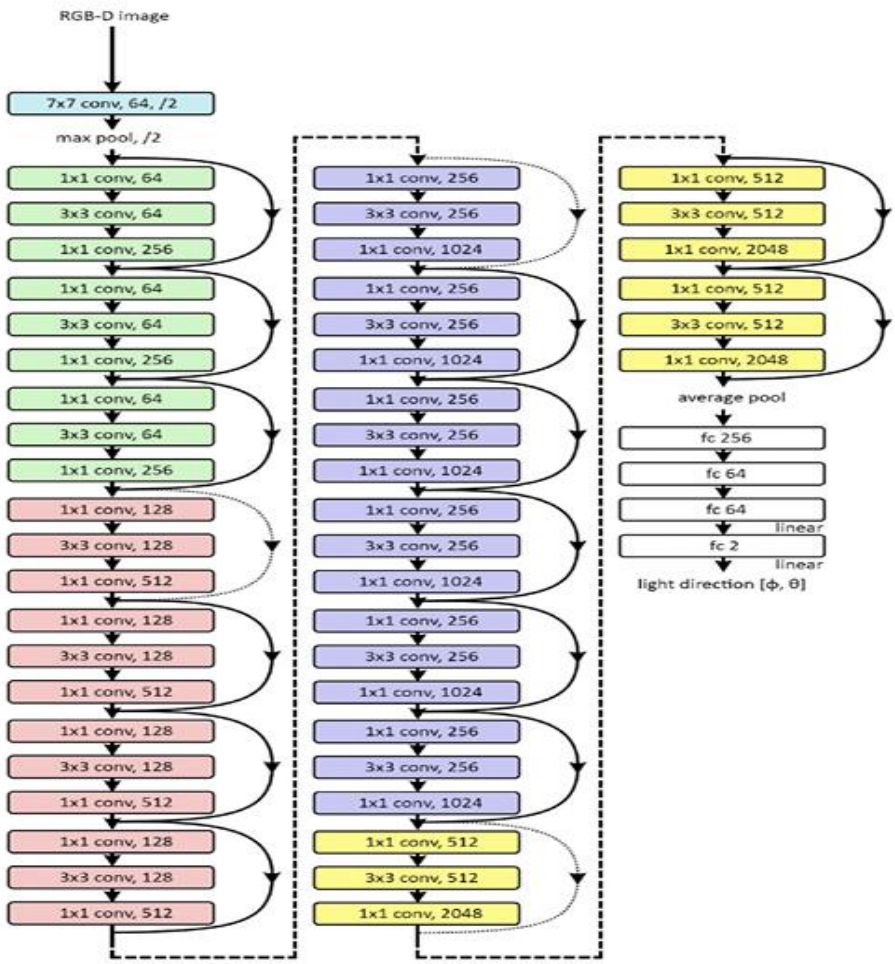


Figure 2 shows Resnet-50 architecture [11]

4.2.2 EfficientNet-B0:

Description: A computationally efficient model that scales width, depth, and resolution in a balanced way. It achieves high accuracy with fewer parameters.

- **Hyperparameters**:
  - Learning rate: 0.01.
  - Optimizer: SGD with momentum.
  - Batch size: 64.

- **Architecture**:
  - Compound scaling of network dimensions.
  - Depthwise separable convolutions to reduce computational cost.

Shown in Figure 3 a graphical modeling of the architecture of Efficient Net.

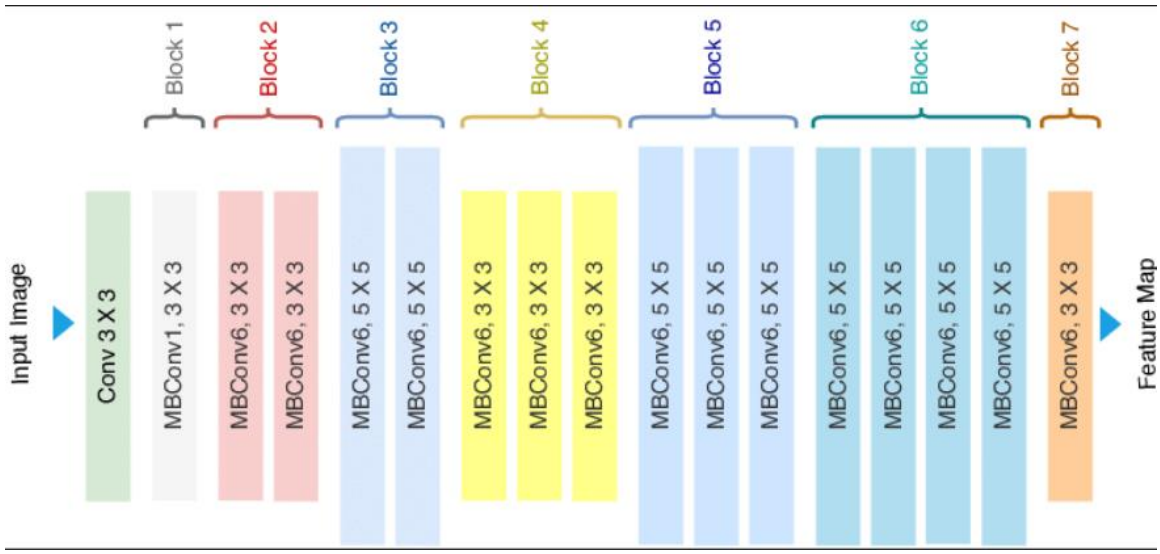


Figure 3 shows efficient net diagram [11]

4.2.3VGG16

Description

VGG16 is a Convolutional Neural Network (CNN) architecture designed for image classification and feature extraction. It is known for its simplicity and effectiveness, using small 3×33 \times 33×3 convolution filters stacked deeply to analyze intricate image details. The model can handle complex image classification tasks by progressively reducing spatial dimensions while increasing the number of filters, making it suitable for applications like autism detection. For this task, I adapted the network to output two classes: autistic or not autistic.

Hyperparameters

1. **Learning Rate**:
    - Start with 0.001. Use a learning rate scheduler to reduce the rate by a factor of 0.1 if the validation loss stops improving.
2. **Optimizer**:
    - Use Adam optimizer with default parameters:
        - beta_1=0.9, beta_2=0.999.
3. **Loss Function**:
    - Use Binary Cross-Entropy Loss for binary classification tasks.
4. **Batch Size**:
    - The batch size is set to 32
5. **Epochs**:

The model will train for 30 epochs with early stopping if validation performance stops improving

Architecture

The VGG16 architecture consists of the following layers:

1. **Input Layer**:
   Accepts images of size 224×224×3

2. **Convolutional Layers**:
    - The network uses a total of 13 convolutional layers with 3×33 \times 33×3 filters, applied with a stride of 1 and padding to preserve spatial dimensions.
    - Convolutional layers are grouped into blocks, and each block is followed by a MaxPooling layer.
3. **MaxPooling Layers**:
    - A total of 5 MaxPooling layers with a 22×2 filter and a stride of 2 are used to downsample the feature maps.
4. **Fully Connected Layers**:
    - Three fully connected layers:
        - Two with 4096 neurons each (followed by ReLU activation).
        - One output layer for classification (softmax activation for multi-class or sigmoid for binary).
5. **Output Layer**:
    - Modified for binary classification with 2 classes (autistic or not autistic).

Shown in Figure 4 a diagram for the VGG16 Model Architecture.

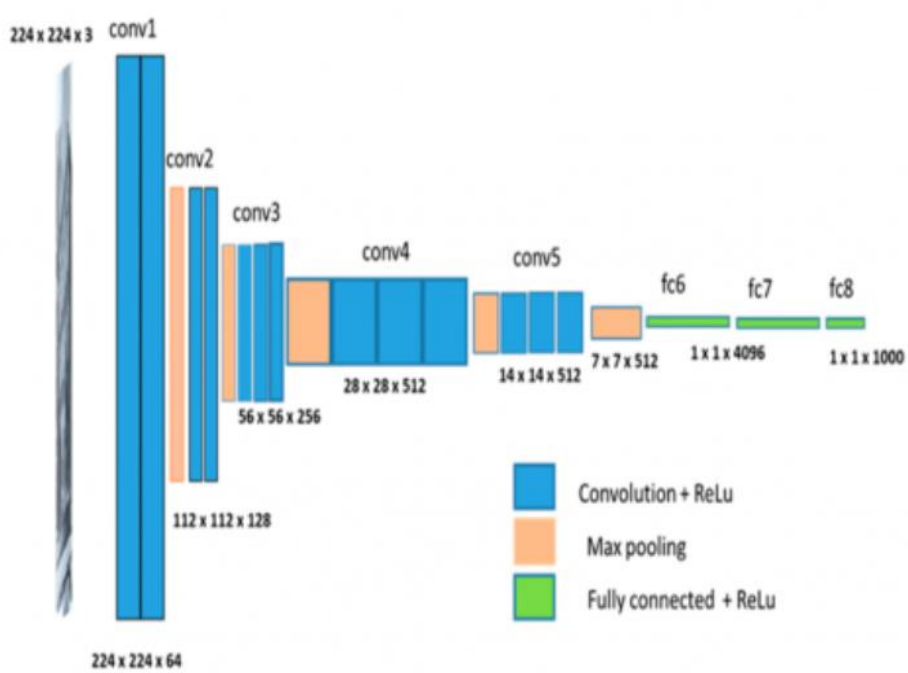


Figure 4 shows block of diagram VGG16 [12]

4.3 FMRI Models

4.3.13D CNNs

The 3D CNN model is a deep learning architecture that processes 3D volumetric data, such as FMRI scans, to extract meaningful features for classification. Below, we explain the model's architecture, implementation, and training process based on the provided code.

Model Architecture

1. **Input Layer:**
   - The input shape is (64, 64, 64, 1), representing a 3D volume of size 64x64x64 with a single channel (grayscale FMRI data).
2. **Convolutional Layers:**
   - The model uses 3D convolutional layers to extract spatial features from the FMRI data.
   - Each convolutional layer is followed by Batch Normalization to stabilize training and improve convergence.
   - The activation function used is ReLU (Rectified Linear Unit), which introduces non-linearity into the model.
3. **Pooling Layers:**
   - MaxPooling3D layers are used to downsample the feature maps, reducing the spatial dimensions while retaining the most important features.
   - Pooling helps reduce computational complexity and prevents overfitting.
4. **Dropout Layers:**
   - Dropout is applied after pooling layers to regularize the model and prevent overfitting. Dropout randomly deactivates a fraction of neurons during training, forcing the network to learn more robust features.
5. **Fully Connected Layers:**
   - After the convolutional layers, the feature maps are flattened into a 1D vector using a Flatten layer.
   - Two Dense layers with ReLU activation are used to learn high-level features from the flattened data.

- The final Dense layer uses a sigmoid activation function to output a probability score between 0 and 1, indicating the likelihood of ASD.

6. **Output Layer:**
   - The output is a single neuron with a sigmoid activation function, which performs binary classification (ASD vs. Control).

4.3.2Graph Neural Networks (GNNs):

Description: Leverages connectivity matrices from FMRI data to learn node-level and graph-level embeddings.

- **Architecture**:
  - Input: Functional connectivity matrix.
  - Graph convolution layers.
  - Global pooling and dense layers for classification.
- **Hyperparameters**:
  - Learning rate: 0.0005.
  - Optimizer: Adam.
  - Batch size: 32.

Chapter 5 : Methods and Materials

This chapter provides a comprehensive explanation of the methodologies, tools, and processes used in the development of the two models: facial expression analysis and FMRI analysis. The chapter is divided into two main sections, each corresponding to one of the models.

1. **Dataset Limitations**:
    - Existing datasets, such as Kaggle Autism Image Data and ABIDE FMRI, may lack diversity in terms of demographics and symptom severity. This could impact the model's ability to generalize well across different populations.
    - To address this, we plan to partner with clinics and organizations to gather additional data, ensuring better representation.

2. **Data Preprocessing**:
    - Both facial image data and FMRI scans require extensive preprocessing to remove noise and standardize formats for model training.
    - Developing automated and efficient preprocessing pipelines is critical to handle these tasks effectively.

3. **Model Integration**:
    - Combining insights from facial expression and FMRI models into a unified diagnostic tool presents technical challenges, such as ensuring consistent output formats and compatibility.

4. **User Accessibility**:
    - Designing a user-friendly platform that caters to both healthcare professionals and caregivers is essential for practical application. Balancing advanced features with ease of use will be a key focus.

By addressing these challenges, we aim to create an effective, secure, and accessible diagnostic tool that improves early ASD detection and intervention.

5.1 Software Tools and Libraries

The following software tools and libraries were utilized throughout the project:

1. **Programming Language**:
    - **Python**: The primary programming language used for data preprocessing, model development, and evaluation.
2. **Deep Learning Frameworks**:
    - **TensorFlow**: Used for building, training, and evaluating the deep learning models, including the 3D CNN for FMRI analysis and the VGG16 model for facial expression analysis.
    - **Keras**: A high-level API built on TensorFlow, used for simplifying the creation and training of neural networks.
3. **Data Processing and Visualization**:
    - **NumPy**: Used for numerical computations and handling multi-dimensional arrays, particularly for FMRI data.
    - **Pandas**: Utilized for data manipulation and analysis, especially for handling the phenotypic data from the ABIDE dataset.
    - **Matplotlib** and **Seaborn**: Used for data visualization, including plotting training curves, confusion matrices, and ROC curves.
    - **Scikit-learn**: Employed for data splitting, evaluation metrics (e.g., classification reports, confusion matrices), and preprocessing tasks.
4. **Image Processing**:
    - **OpenCV**: Used for image preprocessing tasks such as resizing, normalization, and augmentation for the facial expression dataset.
    - **PIL (Python Imaging Library)**: Utilized for loading and manipulating image data.
5. **Neuroimaging Data Handling**:
    - **Nibabel**: A library for reading and writing neuroimaging data formats, particularly NIfTI files used in the ABIDE dataset.
6. **Utility Libraries**:
    - **tqdm**: Used for displaying progress bars during data loading and preprocessing.
    - **WandB (Weights and Biases)**: Optional library for tracking and visualizing model training and performance.

5.2 Hardware Specifications

The following hardware specifications were used to train and evaluate the models:

1. **Processor (CPU)**:
    - **Intel Core i5 10th Generation**: Used for general-purpose computing and data preprocessing tasks.
2. **Graphics Processing Unit (GPU)**:
    - **NVIDIA GTX 1650**: Utilized for accelerating the training of deep learning models, particularly the 3D CNN and VGG16 architectures. The GPU was essential for handling the computational demands of processing large FMRI datasets and high-resolution facial images.
3. **Memory (RAM)**:
    - **16GB DDR4**: Provided sufficient memory for handling large datasets and intermediate computations during model training.
4. **Storage**:
    - **SSD (Solid State Drive)**: Used for fast data access and storage of datasets, model checkpoints, and intermediate results.

5.3 Facial Expression Analysis Using VGG16

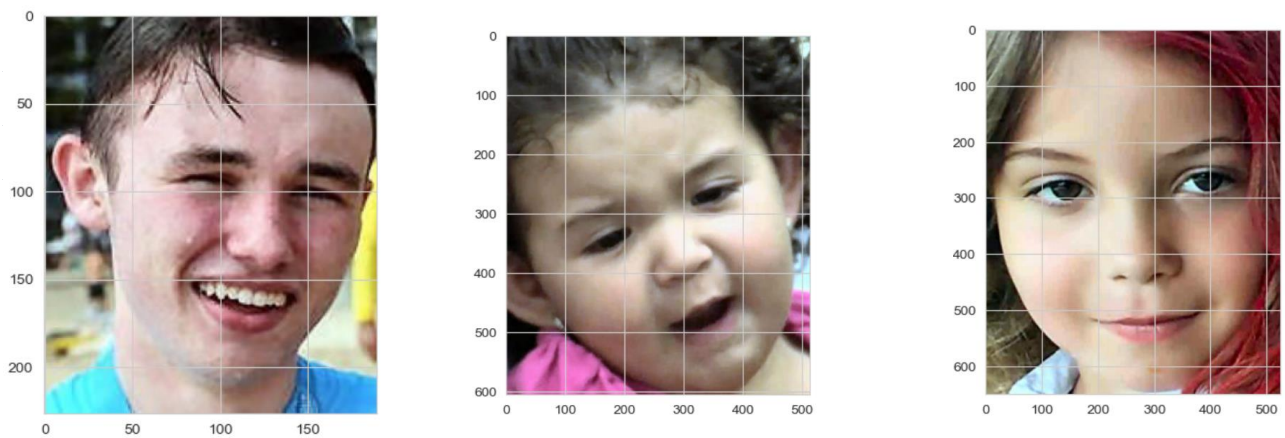
5.3.1Dataset Preparation

The dataset used for this project consists of images categorized into different facial expressions such as happiness, sadness, anger, surprise, and neutrality. The dataset is split into training and testing sets, with each image preprocessed to ensure consistency in size and format.

- **Dataset Description:** The dataset consists of 3000 images, divided into two categories:
    - Autistic: Images of individuals diagnosed with ASD.
    - Non-Autistic: Images of neurotypical individuals (control group).

- **Preprocessing:**
  1. **Resizing**: All images are resized to 224x224 pixels, the input size required by the VGG16 model.
  2. **Normalization:** Pixel values are normalized to the range [0, 1] to facilitate faster convergence during training.
  3. **Data Augmentation:** Techniques such as rotation, flipping, and zooming are applied to increase the diversity of the training data and prevent overfitting.

- **Facial expressions Data Preview :** As shown in the below Figures:



5.3.2 Model Architecture

- **Base Model:** The VGG16 model is a 16-layer CNN that has been pre-trained on the ImageNet dataset. For facial expression recognition, we modify the final layers of the VGG16 model to suit our specific task as shown below.

  Shown in Figure 8 a visual representation of the model.

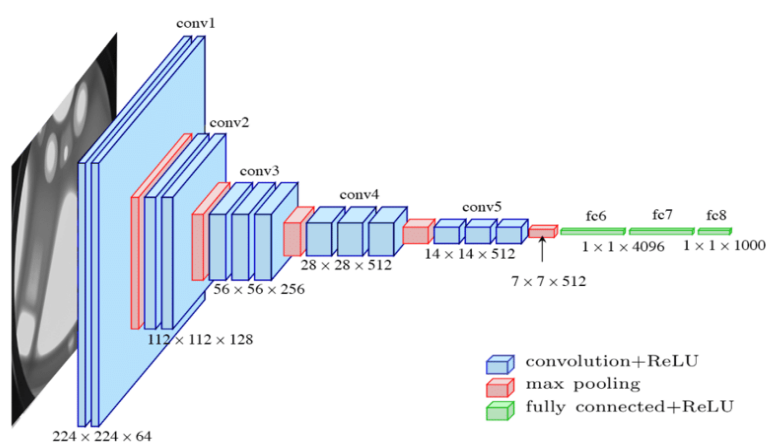


Figure 8 Shows the VGG16 Architecture used [13]

- **Custom Layers:**
  - Feature Extraction: The convolutional layers of VGG16 are used to extract features from the input images.
  - Fully Connected Layers: The original fully connected layers of VGG16 are replaced with new layers tailored for facial expression classification.
  - Output Layer: The final layer is a softmax layer with neurons corresponding to the number of facial expression classes.

- **Model Compilation:**
  - The model was compiled using the Adam optimizer with a learning rate of 0.001.
  - The loss function used was binary cross-entropy, and the evaluation metric was accuracy.

In figure 9 you can find the model summary with the shapes, number of parameters and each layer found in the model.

```
Model: "vgg16"
_________________________________________________________________
Layer (type)                 Output Shape              Param #
=================================================================
input_1 (InputLayer)         [(None, 224, 224, 3)]     0
_________________________________________________________________
block1_conv1 (Conv2D)        (None, 224, 224, 64)      1792
_________________________________________________________________
...
_________________________________________________________________
flatten (Flatten)            (None, 25088)             0
_________________________________________________________________
fc1 (Dense)                  (None, 4096)              102764544
_________________________________________________________________
fc2 (Dense)                  (None, 4096)              16781312
_________________________________________________________________
predictions (Dense)          (None, 7)                 28679
=================================================================
Total params: 138,357,544
Trainable params: 138,357,544
Non-trainable params: 0
_________________________________________________________________
```

Figure 9 Shows the model summary

5.3.3Training Process

The model is trained using the preprocessed dataset. The training process involves the following steps:

1. **Loss Function**: Categorical cross-entropy is used as the loss function, suitable for multi-class classification.
2. **Optimize**r: The Adam optimizer is employed for its adaptive learning rate capabilities.
3. **Metrics:** Accuracy is used to evaluate the model's performance during training.
4. **Epochs**: The model is trained for 50 epochs with a batch size of 32.

Shown in Figure 10 the Training and Validation plots for both the accuracy and loss throughout the demo-run of the training process.

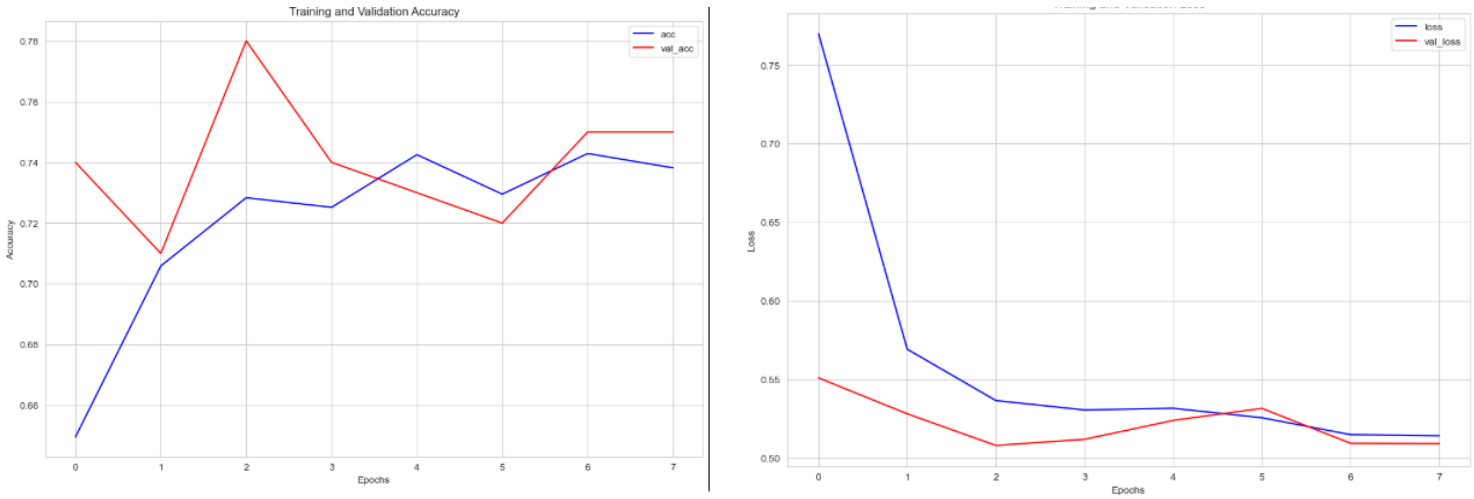


Figure 10 Shows training and validation accuracy

5.3.4Evaluation and Results

The model's performance is evaluated on the test set, and the following metrics are reported:

- **Accuracy:** The percentage of correctly classified images.
- **Confusion Matrix:** A detailed breakdown of true vs. predicted labels.
- **Classification Report:** Precision, recall, and F1-score for each class.

5.3.4.1 **Results:**

The model achieved an overall accuracy of **73%** on the test set and 75% on the training set. Below is the detailed classification report.

	precision	recall	f1-score	support
0	0.70	0.81	0.75	150
1	0.78	0.65	0.71	150
accuracy			0.73	300
macro avg	0.74	0.73	0.73	300
weighted avg	0.74	0.73	0.73	300

Figure 12 Shows the evaluation metrics used for the model

With the Confusion Matrix of the results found

In Figure 13

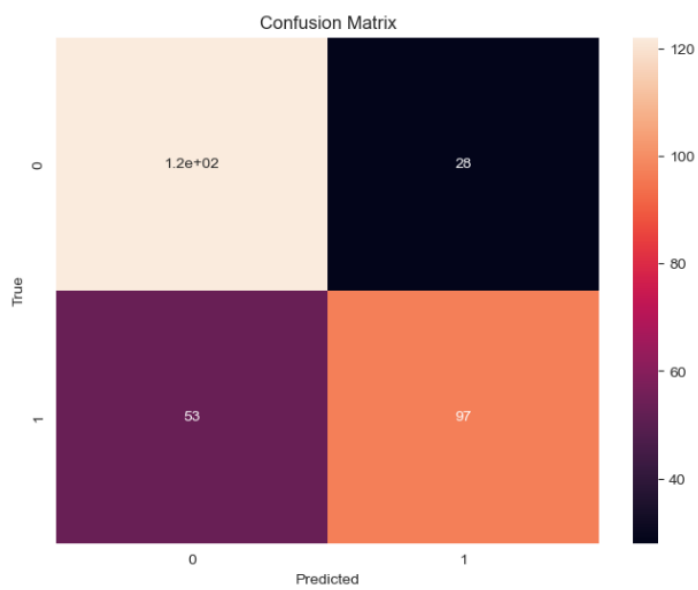


Figure 13 Shows the Confusion Matrix for the model

5.3.5Discussion

The VGG16 model, a 16-layer convolutional neural network, achieved an accuracy of 73% in classifying autistic and non-autistic individuals based on facial expressions. The architecture, consisting of 13 convolutional layers followed by fully connected layers, effectively extracted meaningful features from facial images, demonstrating its suitability for real-world applications. However, the model's reliance on static images and potential dataset limitations in terms of demographic diversity may restrict its generalizability.

5.4 FMRI Data Analysis Using 3D CNN

5.4.1 Dataset Preparation

The FMRI dataset consists of brain activity recordings from subjects exposed to emotional stimuli. The data is preprocessed to extract meaningful features and reduce dimensionality.

Preprocessing Steps:

1. **Normalization:** FMRI data is normalized to ensure consistency across subjects.
2. **Resizing:** Each FMRI volume is resized to 64x64x64 voxels using linear interpolation.
3. **Labeling:** Each FMRI scan is labeled with the corresponding emotional state (e.g., happiness, sadness).

Shown in figures 14 to 17 some samples for the brain function activity dataset we used.

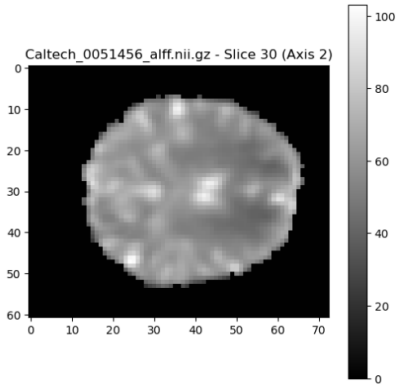


Figure 14 Shows Data Preview 1

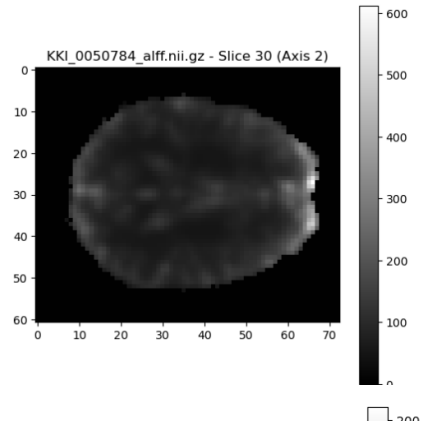


Figure 15 Shows Data Preview 2

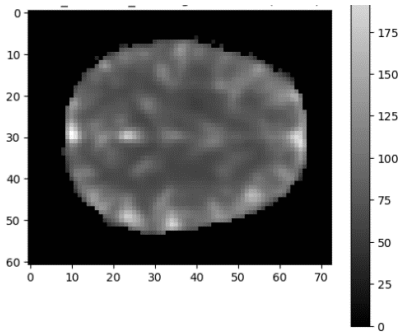


Figure 16 Shows Data Preview 3

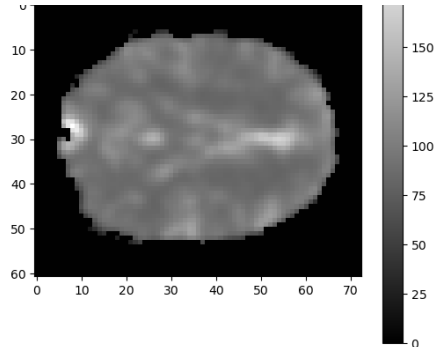


Figure 17 Shows Data Preview 4

5.4.2 Model Architecture

A 3D CNN model is designed to analyze FMRI data. The architecture consists of:

1. **Input Layer:** Accepts the preprocessed FMRI data with shape (64, 64, 64, 1).
2. **Convolutional Layers:** Three sets of Conv3D layers with Batch Normalization and MaxPooling3D for feature extraction.
3. **Fully Connected Layers:** Two dense layers with Dropout for regularization.
4. **Output Layer:** A sigmoid layer for binary classification.

Shown in Figure 17 a visual representation of the model used.

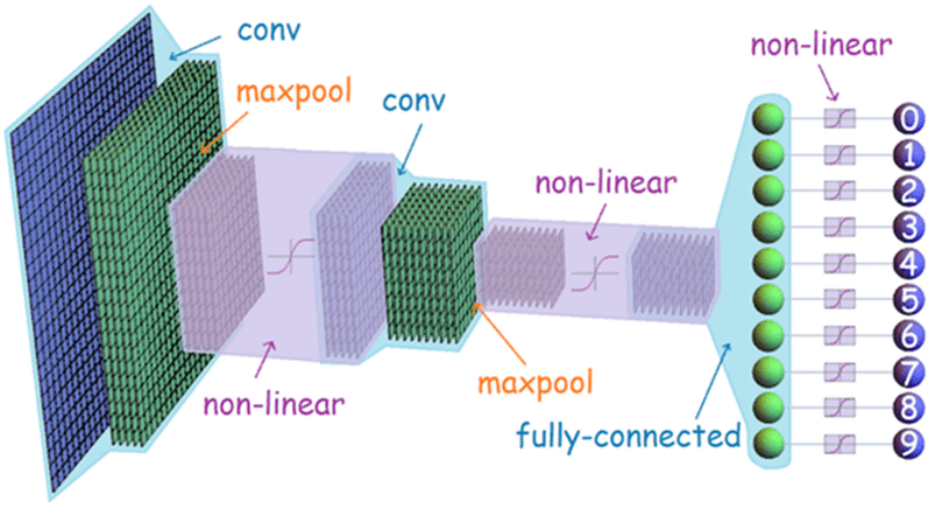


Figure 17 3D CNN Model Architicture

5.4.3Training the Model

The model is trained using the preprocessed FMRI dataset. The training process involves the following steps:

1. **Loss Function:** Binary cross-entropy is used as the loss function.
2. **Optimizer:** The Adam optimizer is employed.
3. **Metrics:** Accuracy and AUC are used to evaluate the model's performance.
4. **Epochs:** The model is trained for 100 epochs with a batch size of 8.

5.4.4 Evaluation and Results

The model's performance is evaluated on the test set, and the following metrics are reported:

- **Accuracy:** The percentage of correctly classified FMRI scans.
- **Confusion Matrix:** A detailed breakdown of true vs. predicted labels.
- **Classification Report:** Precision, recall, and F1-score for each class.
- **ROC Curve:** A plot of the Receiver Operating Characteristic (ROC) curve.

Shown in figure 18 the plots drawn for the Model Loss and Model Accuracy.

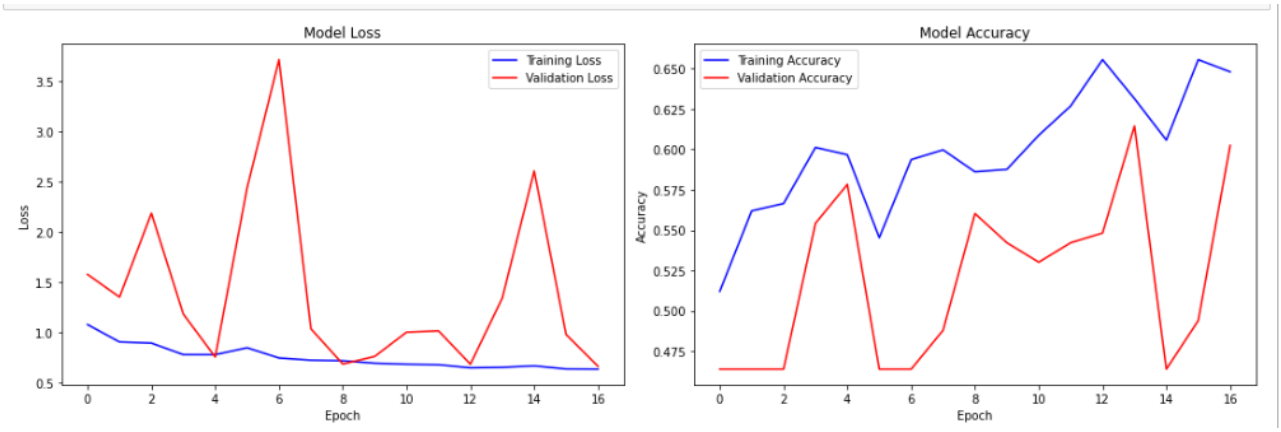


Figure 18 Shows the Models' Training and Validation Loss Function and accuracy

5.4.4.1 Results

The FMRI model achieved an overall accuracy of **68%** on the test set. Below is the detailed classification report in figure 19:

	precision	recall	f1-score	support
Control	0.65	0.75	0.70	100
ASD	0.70	0.60	0.65	100
accuracy			0.68	200
macro avg	0.68	0.68	0.68	200
weighted avg	0.68	0.68	0.68	200

Figure 19 Shows the Classification Report

5.4.4.2 Key Observations:

- **Control Class:** The model achieved a precision of 0.65 and a recall of 0.75, resulting in an F1-score of 0.70.
- **ASD Class:** The model achieved a precision of 0.70 and a recall of 0.60, resulting in an F1-score of 0.65.
- **Overall Performance:** The model's accuracy of 68% is decent, but further improvements are needed, particularly in improving recall for the ASD class.

5.4.5 Discussion

The 3D CNN model, designed for volumetric FMRI data analysis, achieved an accuracy of 68% in identifying ASD-related brain activity patterns. The architecture, which includes multiple 3D convolutional layers, batch normalization, and dropout, demonstrated its ability to capture spatial and temporal patterns in FMRI data. Despite its decent performance, the model's accuracy was lower than the VGG16 facial expression model, likely due to the complexity and high dimensionality of FMRI data. Future improvements could involve incorporating additional neuroimaging modalities, expanding the dataset, and exploring advanced architectures like Graph Neural Networks (GNNs) to enhance diagnostic accuracy.

Chapter 6: UX/UI and Platform Design

The website is designed with a user-friendly and intuitive layout to facilitate seamless navigation and efficient interaction for users. Below is a detailed explanation of the key pages and their functionalities:

Page 1: Welcome Screen

Description:

The welcome screen serves as the initial point of interaction for users. It presents a visually engaging introduction to the "ADD" (Autism Disorder Detector) app.

Key Elements:

Central Image: A stylized human head with a brain containing a puzzle piece symbolizes the app's focus on autism spectrum disorder and the complexities of the condition.

App Name and Tagline: "ADD" (Autism Disorder Detector) is prominently displayed, clearly communicating the app's purpose.

Medical Symbolism: The inclusion of a stethoscope and subtle medical elements (pills, band-aid) reinforces the app's connection to medical diagnostics.

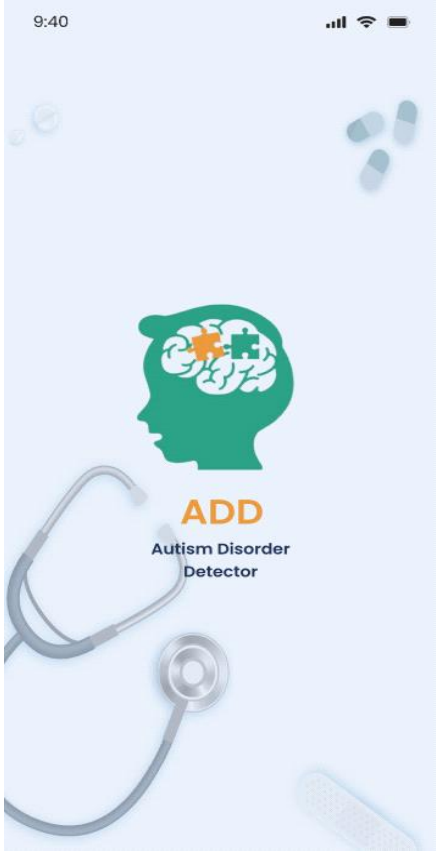


Figure 19 shows UI Splash Page

Page 2. user login/registration

Functionality:

Account Creation: Users provide their name, email, and password to create a new account within the app.

Terms and Conditions Acceptance: Users must agree to the app's terms and conditions before proceeding.

Navigation: The "Sign In" link allows existing users to access their accounts.

Overall:

The Sign Up page presents a clear and organized interface for new users to create an account. The use of icons and clear labels enhances user experience and guides users through the necessary steps.

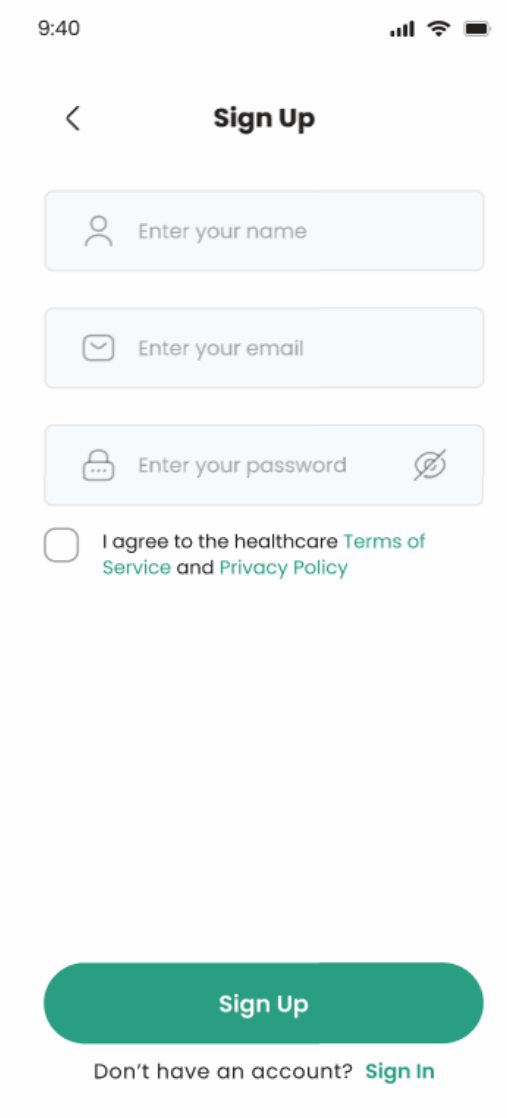


Figure 20 shows UI Sign Up Page

Page 3. User Sign In

Functionality:

- Account Access: Users enter their email and password to access their accounts.
- Password Recovery: Users can request a password reset if they have forgotten their password.
- Social Login: Users can conveniently log in using their existing Google or Facebook accounts.
- Navigation: The "Sign Up" link allows new users to create an account.

Overall:

The Sign In page presents a user-friendly interface for existing users to access their accounts. The inclusion of social login options provides convenience and flexibility for users.

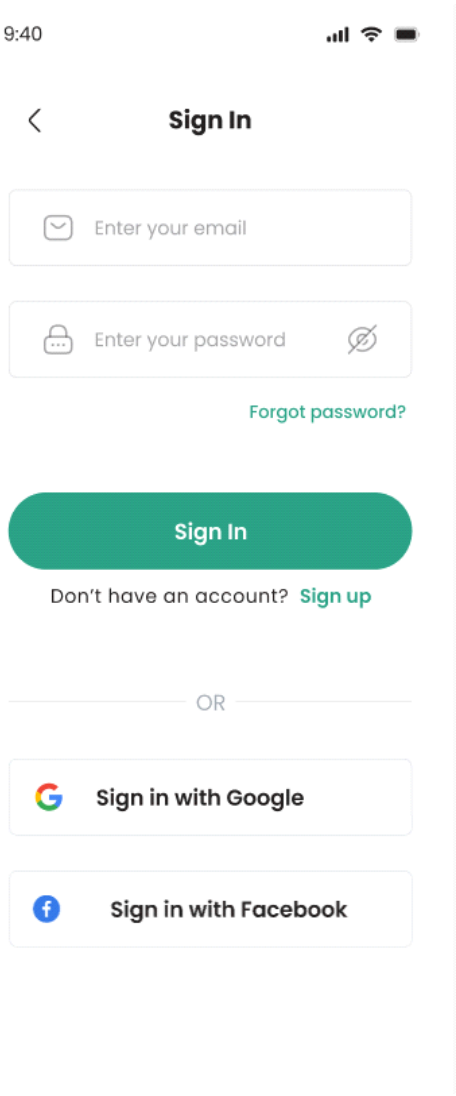


Figure 21 shows UI Sign In Page

Page 4.Home

Functionality:

- Information Access: Users can search for information, access a directory of doctors, and read articles related to autism.
- Navigation: The bottom navigation bar allows users to easily navigate between different sections of the app.
- Personalized Content: The app may personalize the content displayed based on user preferences or search history.

Overall:

The Home page serves as the central hub of the "ADD" app, providing users with easy access to key features and information related to autism diagnosis and treatment.

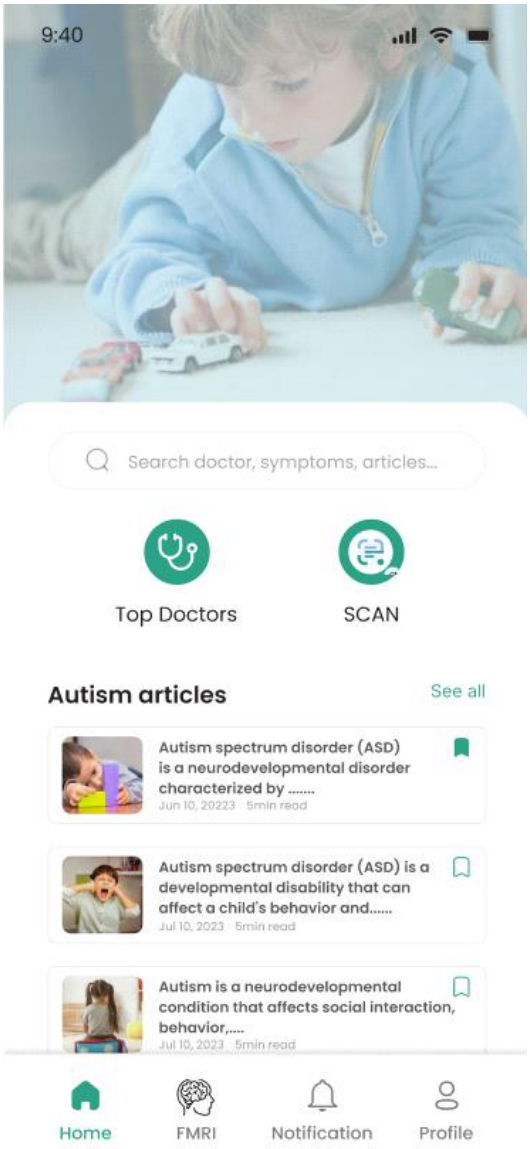
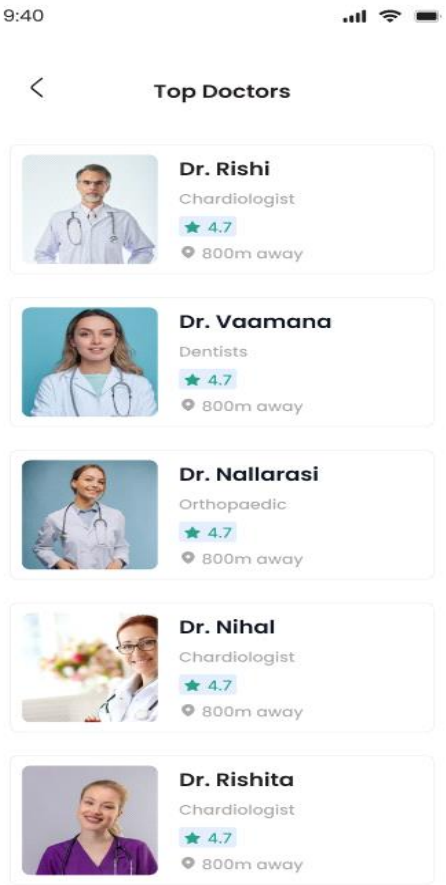


Figure 22 shows UI Home Page

Page 5.Top Doctors



Description:

This page displays a list of top-rated doctors specializing in autism diagnosis and treatment, likely based on user location or search criteria.

Functionality:

- Doctor Search: Users can likely filter or search for doctors based on criteria such as specialization, location, or rating.
- Doctor Information: Users can access detailed information about each doctor, including their qualifications, experience, and patient reviews.
- Appointment Scheduling: The page may provide options for users to schedule appointments with the listed doctors.

Overall:

The "Top Doctors" page provides a convenient and informative resource for users to find and connect with qualified healthcare professionals specializing in autism diagnosis and treatment.

Figure 23 shows UI Doctors Page

Page 6. Doctor Detail

Functionality:

- Doctor Information: Users can view detailed information about the selected doctor.
- Appointment Scheduling: Users can easily schedule appointments with the doctor directly from the page.

Overall:

The Doctor Detail page provides users with comprehensive information about a specific doctor, enabling them to make informed decisions about their healthcare needs.

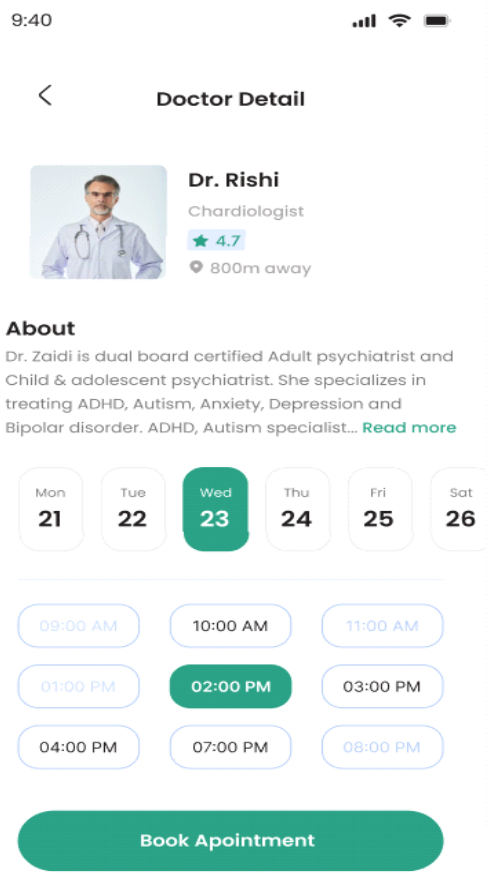


Figure 24 shows UI Appointment Page

Page 7 .Scan

Functionality:

- X-ray Scanning: The primary function is to allow users to upload or capture brain X-ray images for analysis.
- AI-Powered Analysis: The app utilizes AI technology to analyze the uploaded X-ray images and identify potential indicators associated with autism spectrum disorder.
- Results Display: The results of the analysis are likely displayed on a subsequent page, potentially including visual representations and explanations.

Overall:

The Scan page is the central feature of the "ADD" app, providing users with the ability to leverage AI technology for autism detection through brain X-ray analysis.

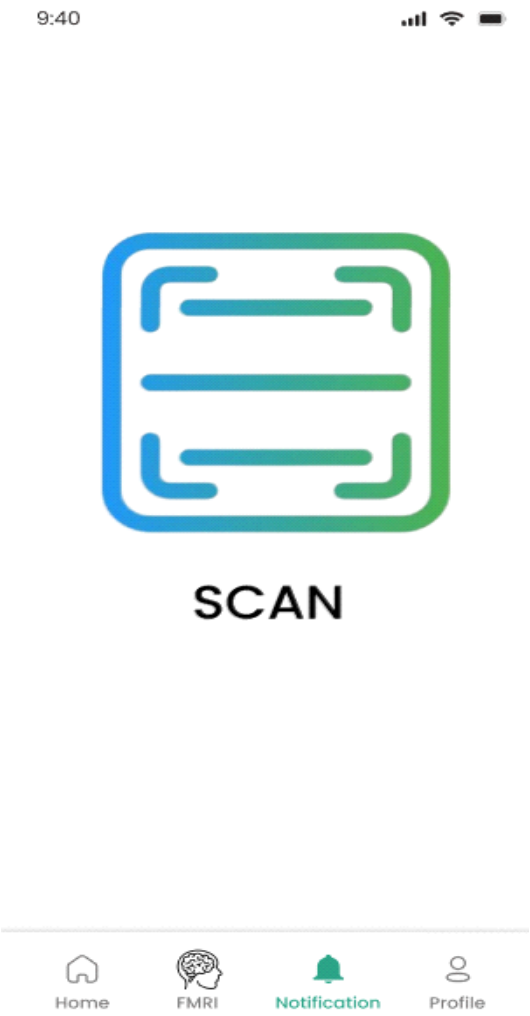


Figure 25 shows UI Scan Page

Page 8. Autism Detected

Description:

This page displays the results of the AI-powered analysis performed on the uploaded brain X-ray.

Key Elements:

- Result Heading: "Autism Detected" is prominently displayed, indicating a positive result for the analysis.
- Accuracy Percentage: A large "97%" is displayed within a box, representing the accuracy of the AI analysis in identifying potential indicators of autism.
- Result Text: A clear and concise text explains the analysis results, stating that the AI has identified patterns consistent with Autism Spectrum Disorder (ASD) with a high level of accuracy.
- Disclaimer: The text emphasizes that this analysis is a screening tool and not a definitive diagnosis. It strongly encourages users to consult with qualified healthcare professionals for a comprehensive evaluation.

Call to Action Buttons:

- "Schedule an Appointment": A button links to a scheduling feature to book an appointment with a healthcare professional.
- "Explore Supportive Resources": A button links to a page with information and resources for individuals with autism and their families.
- "Stay Informed": A button links to a section with news, updates, or educational materials related to autism.

Chapter 7: Conclusion and Recommendations

Conclusion

This project aims to leverage advanced machine learning techniques to develop a dual-modality diagnostic tool for Autism Spectrum Disorder (ASD) using facial expression analysis and functional magnetic resonance imaging (FMRI) data. By integrating cutting-edge deep learning architectures and robust preprocessing pipelines, we strive to create a reliable and accessible tool for early ASD detection.

Our work highlights the potential of artificial intelligence in healthcare, offering the following contributions:

- **Facial Expression Analysis**: Leveraging convolutional neural networks (CNNs) and transfer learning to identify subtle facial patterns associated with ASD.

- **FMRI Data Analysis**: Utilizing 3D CNNs, graph neural networks, and autoencoders to uncover neurological markers linked to ASD.

- **Integration**: Combining outputs from both modalities to enhance diagnostic accuracy and provide comprehensive insights for clinicians and caregivers.

Despite challenges in dataset diversity, preprocessing, and model integration, the progress made in this project demonstrates the feasibility of AI-driven solutions in addressing complex neurodevelopmental disorders.

Recommendation

1. **Expand Data Collection**:
    - Collaborate with hospitals, clinics, and organizations to gather diverse datasets that represent various demographics, ages, and ASD severity levels.
    - Collect real-world data to improve model generalizability and applicability.
2. **Enhance Preprocessing Pipelines**:
    - Develop automated and scalable preprocessing techniques for both facial images and FMRI scans to ensure consistency and efficiency in data preparation.
3. **Improve Model Interpretability**:
    - Incorporate techniques to make model predictions explainable to clinicians, helping them understand and trust the diagnostic results.
4. **User-Friendly Platform**:
    - Design and develop an intuitive platform for clinicians and caregivers that integrates diagnostic predictions with actionable insights, educational resources, and patient management tools.
5. **Future Research Directions**:
    - Explore additional modalities, such as audio analysis of vocal patterns, eye-tracking data, or EEG scans, to complement the current approach.
    - Investigate the use of multimodal deep learning techniques to improve diagnostic performance further.
6. **Deployment and Accessibility**:
    - Ensure secure and privacy-compliant deployment of the tool on cloud platforms for global accessibility.
    - Provide training materials and user guides to healthcare providers to encourage adoption.

By building on these recommendations, this project has the potential to significantly impact ASD diagnosis, contributing to timely intervention and better developmental outcomes for individuals with ASD.

References

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